

Volatility forecasting - HAR / HARQ / GBM / GBME(GBM+Earning) / GBME+graph (all metrics, DM-bolded)

GBM = gamma-loss Gradient Boosting Machine (HistGradientBoostingRegressor) on own-history features; GBME(GBM+Earning) = GBM + External earnings features; GBME+graph adds the cross-firm correlation-graph neighbour aggregate. QLIKE is the primary loss (equal to the gamma deviance up to a constant); Diebold-Mariano tests use per-trading-date mean loss.

S&P 500

MSE x1e-7 - RMSE/MAE x1e-4 - QLIKE unscaled. **Bold** on MSE/RMSE/MAE = numerically best (lowest) value in that horizon; **bold** on QLIKE = significantly best by date-clustered Diebold-Mariano (p<0.05).

h	Model	MSE	RMSE	MAE	QLIKE
1	HAR	5.754	7.586	2.132	0.3697
1	HARQ	5.648	7.515	2.151	0.3780
1	GBM	5.557	7.454	2.089	0.3570
1	GBME(GBM+Earning)	5.397	7.346	2.017	0.3172
1	GBME+graph	5.424	7.365	2.010	0.3158
5	HAR	6.141	7.836	2.297	0.4364
5	HARQ	6.132	7.831	2.309	0.4377
5	GBM	6.117	7.821	2.255	0.4116
5	GBME(GBM+Earning)	5.972	7.728	2.180	0.3705
5	GBME+graph	6.063	7.787	2.196	0.3715
10	HAR	6.214	7.883	2.356	0.4615
10	HARQ	6.205	7.877	2.358	0.4614
10	GBM	6.262	7.913	2.312	0.4307
10	GBME(GBM+Earning)	6.118	7.822	2.239	0.3912
10	GBME+graph	6.240	7.899	2.272	0.3939
22	HAR	6.327	7.954	2.419	0.4855
22	HARQ	6.311	7.944	2.415	0.4831
22	GBM	6.233	7.895	2.359	0.4464
22	GBME(GBM+Earning)	6.098	7.809	2.277	0.4074
22	GBME+graph	6.087	7.802	2.300	0.4110

DM (QLIKE): GBME vs GBM h1 p=0.000 - GBME vs GBM h5 p=0.000 - GBME vs GBM h10 p=0.000 - GBME vs GBM h22 p=0.000 (all significant → GBME QLIKE bolded).

HOSE

MSE x1e-7 - RMSE/MAE x1e-4 - QLIKE unscaled. **Bold** on MSE/RMSE/MAE = numerically best (lowest) value in that horizon; **bold** on QLIKE = significantly best by date-clustered Diebold-Mariano (p<0.05).

h	Model	MSE	RMSE	MAE	QLIKE
1	HAR	6.604	8.126	5.193	1.8117
1	HARQ	6.389	7.993	5.062	1.7854
1	GBM	5.399	7.348	3.944	1.5679
1	GBME(GBM+Earning)	5.477	7.401	3.966	1.5730
1	GBME+graph	5.555	7.453	3.980	1.5783

h	Model	MSE	RMSE	MAE	QLIKE
5	HAR	6.626	8.140	5.227	1.8218
5	HARQ	6.519	8.074	5.152	1.8075
5	GBM	5.796	7.613	4.265	1.6478
5	GBME(GBM+Earning)	5.795	7.613	4.276	1.6485
5	GBME+graph	5.815	7.625	4.256	1.6494
10	HAR	6.660	8.161	5.250	1.8277
10	HARQ	6.589	8.118	5.198	1.8180
10	GBM	5.966	7.724	4.425	1.6842
10	GBME(GBM+Earning)	5.962	7.721	4.441	1.6862
10	GBME+graph	5.959	7.720	4.407	1.6837
22	HAR	6.716	8.195	5.282	1.8355
22	HARQ	6.677	8.171	5.250	1.8299
22	GBM	6.211	7.881	4.615	1.7263
22	GBME(GBM+Earning)	6.211	7.881	4.627	1.7262
22	GBME+graph	6.207	7.878	4.604	1.7267

DM (QLIKE): GBME vs GBM h1 p=0.000 - GBME vs GBM h5 p=0.214 - GBME vs GBM h10 p=0.122 - GBME vs GBM h22 p=0.919 (earnings does not beat GBM → own-history GBM QLIKE bolded).